

Datasets in scope (D7 excluded — FAIL reads)

Dataset	Organism	Chemistry	Reads	Ref	FAST5
D1	SARS-CoV-2	R9.4	1,383	31 KB	11.00 GB
D2	E. coli	R9.4	89	5 MB	27.00 GB
D3	S. cerevisiae	R9.4	500	12 MB	0.40 GB
D4	C. reinhardtii	R9.4	500	112 MB	2.00 GB
D6	E. coli	R10.4	294	5 MB	98.00 GB
RB_ecoli	E. coli	R10.4.1	12,000	5 MB	0.93 GB
RB_dmel	D. melanogaster	R10.4.1	12,000	143 MB	0.73 GB
RB_hsap_full	H. sapiens (GRCh)	R10.4.1	12,000	3.0 GB	7.10 GB

Datasets — extended characterization

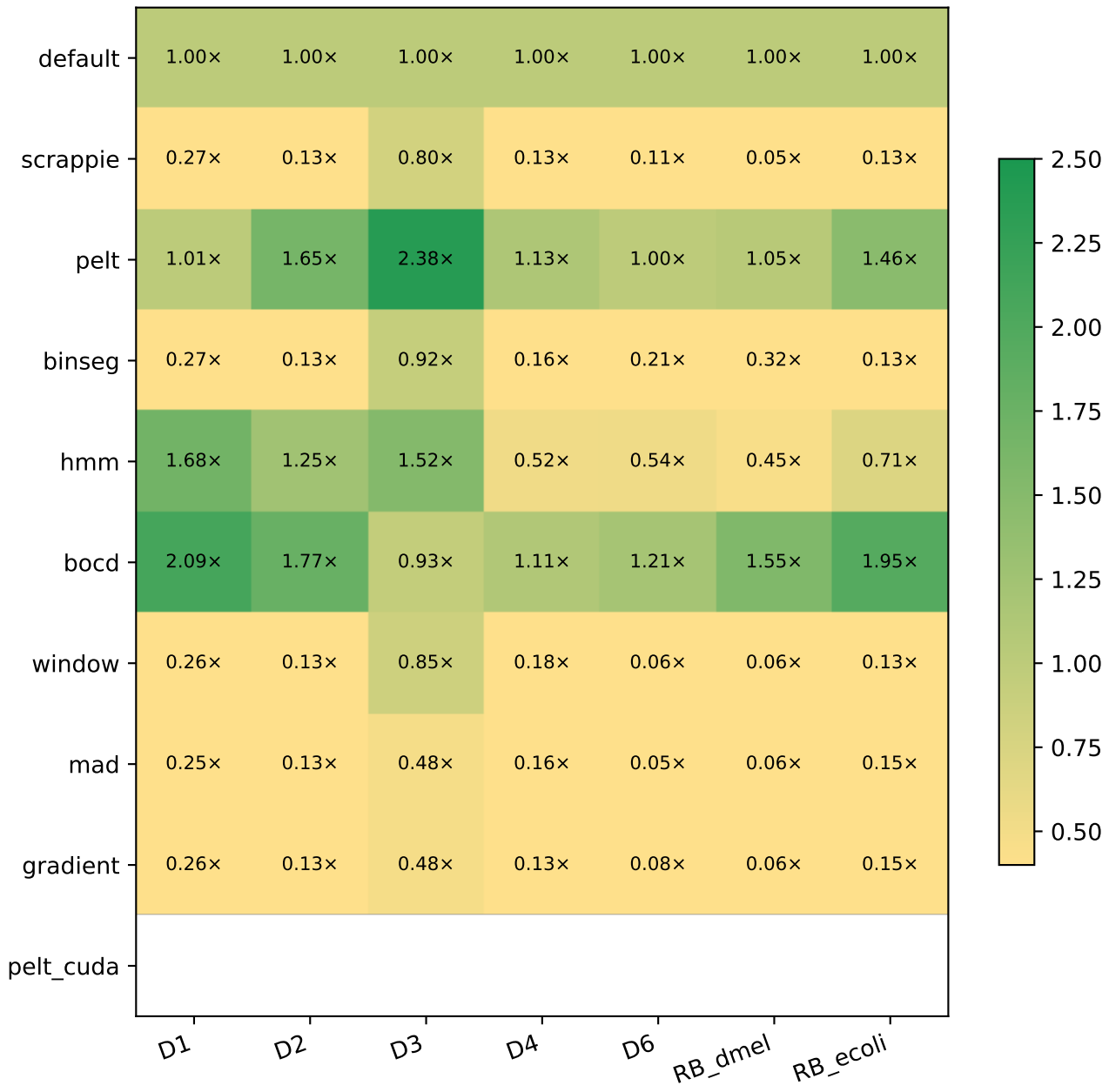
DS	Organism	Chem	Reads	Ref bp	GC%	contigs	N50 (Mb)	N%	masked%
D1	SARS-CoV-2	R9.4	1,383	0.03 Mb	38.0	1	0.03	0.00	0.0
D2	E. coli	R9.4	89	5.23 Mb	50.5	1	5.23	0.00	0.0
D3	S. cerevisiae	R9.4	500	12.16 Mb	38.1	17	0.92	0.00	0.0
D4	C. reinhardtii	R9.4	500	111.10 Mb	64.1	53	7.78	3.65	32.4
D6	E. coli	R10.4	294	5.23 Mb	50.5	1	5.23	0.00	0.0
RB_dmel	D. melanogaster	R10.4.1	12,000	143.73 Mb	42.0	1,870	25.29	0.80	20.9
RB_ecoli	E. coli	R10.4.1	12,000	5.23 Mb	50.5	1	5.23	0.00	0.0

Ref bp = total reference length; *N50* = contig N50; *N%* = ambiguous N bases; *masked%* = soft-masked low-complexity / repeats (lowercase in FASTA).

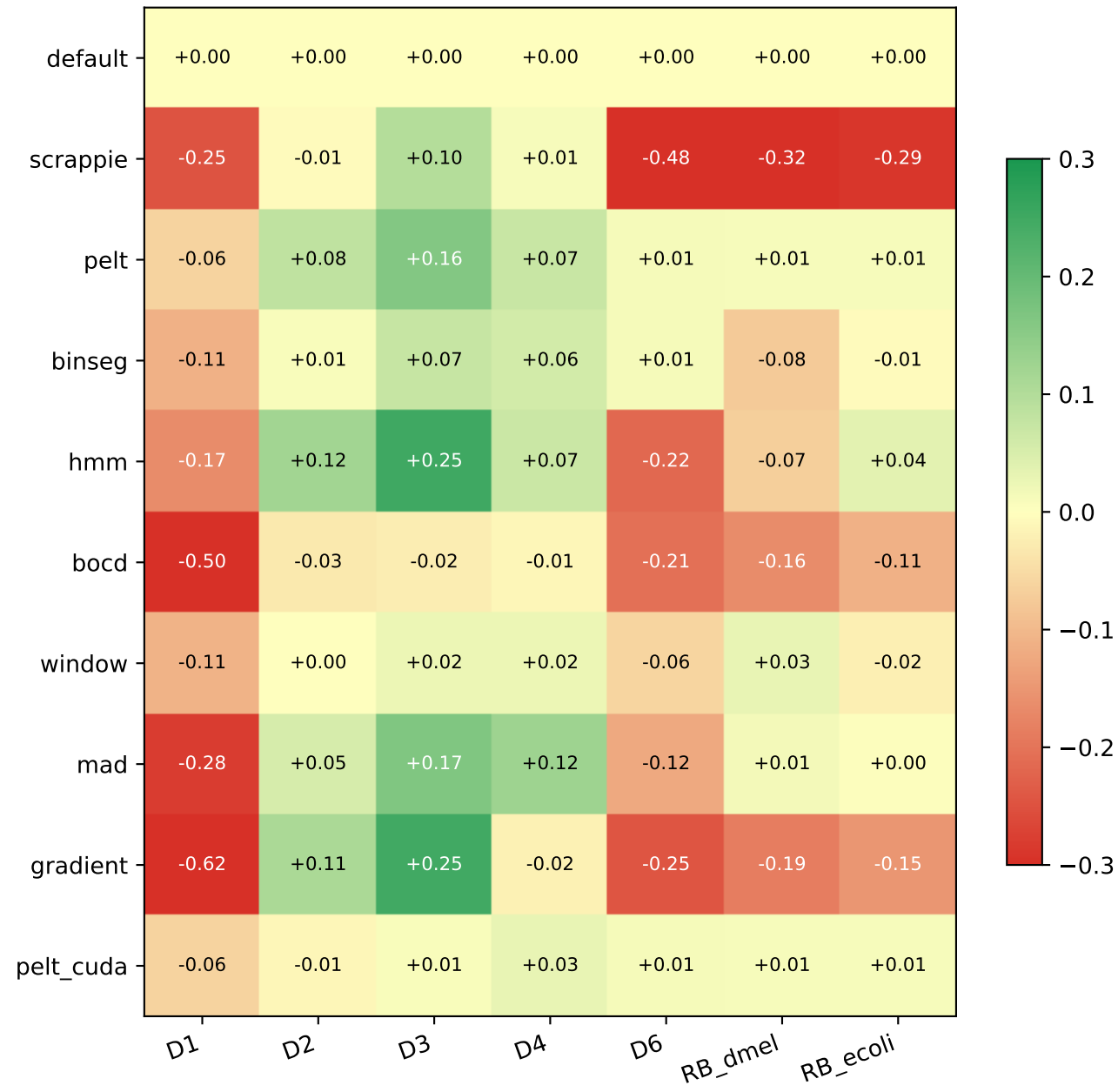
F1 across (segmenter, dataset) — best CV-tuned config used for HMM/BOCD



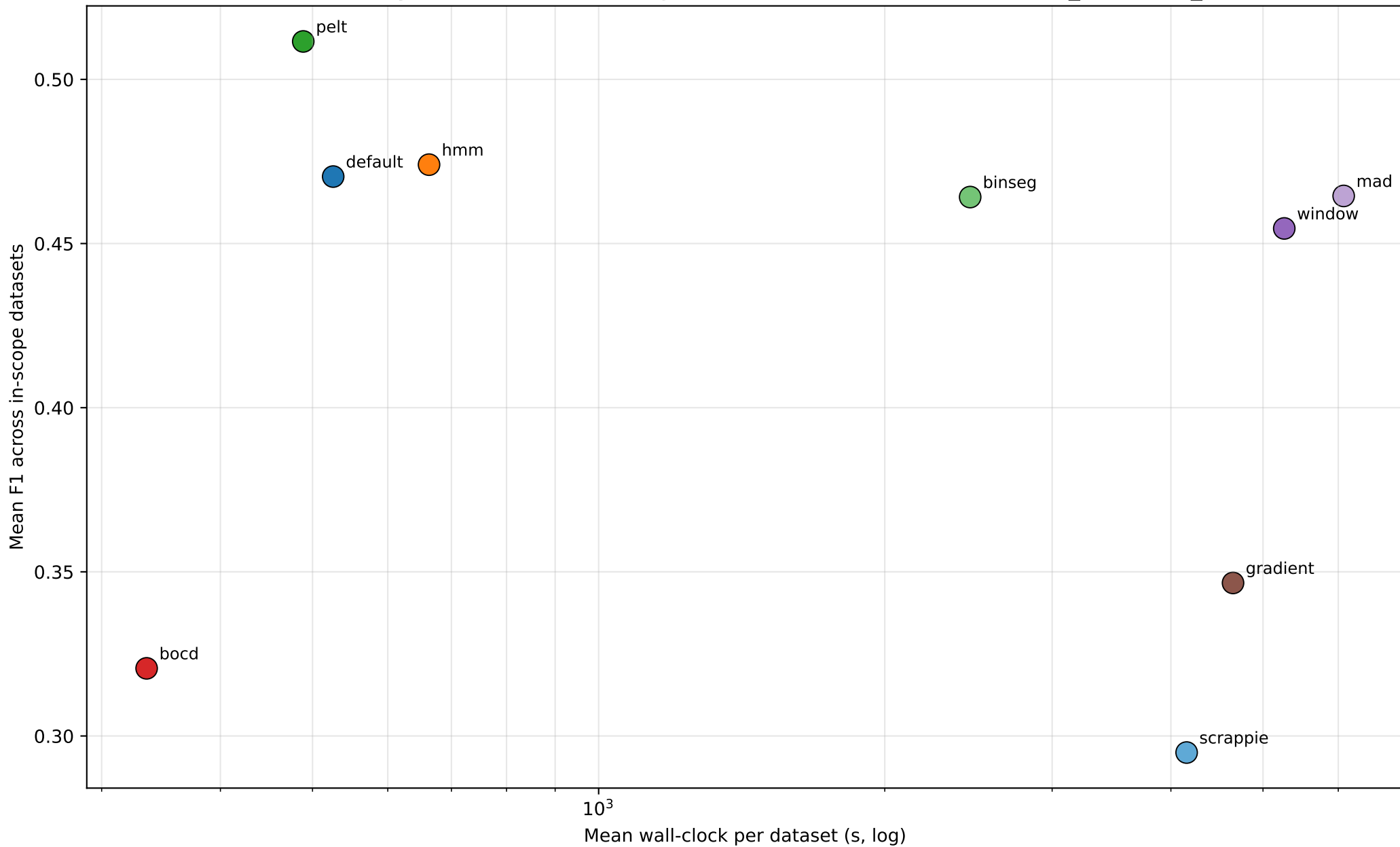
Speedup vs default



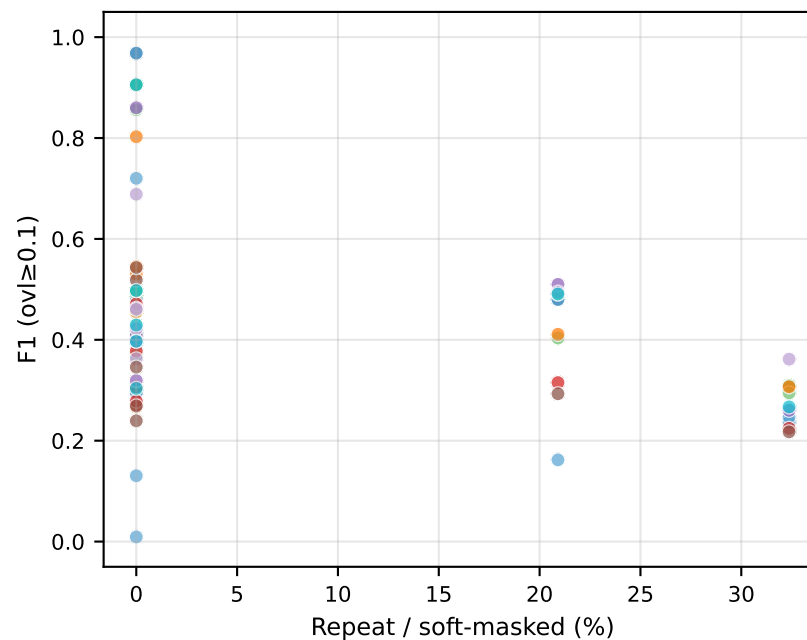
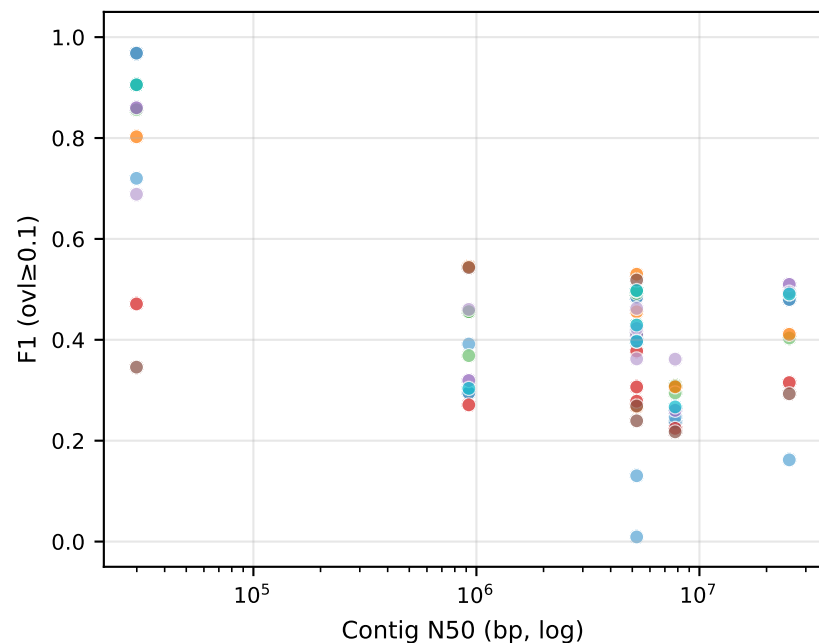
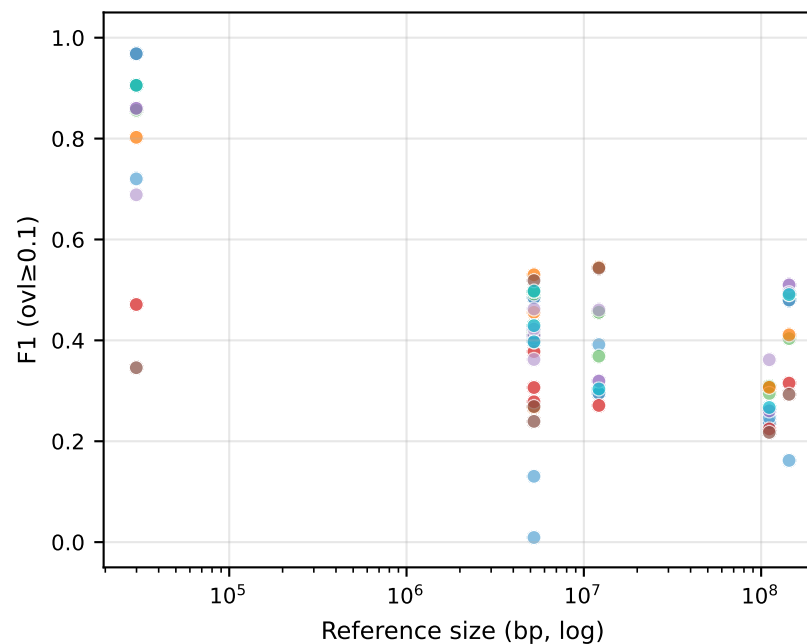
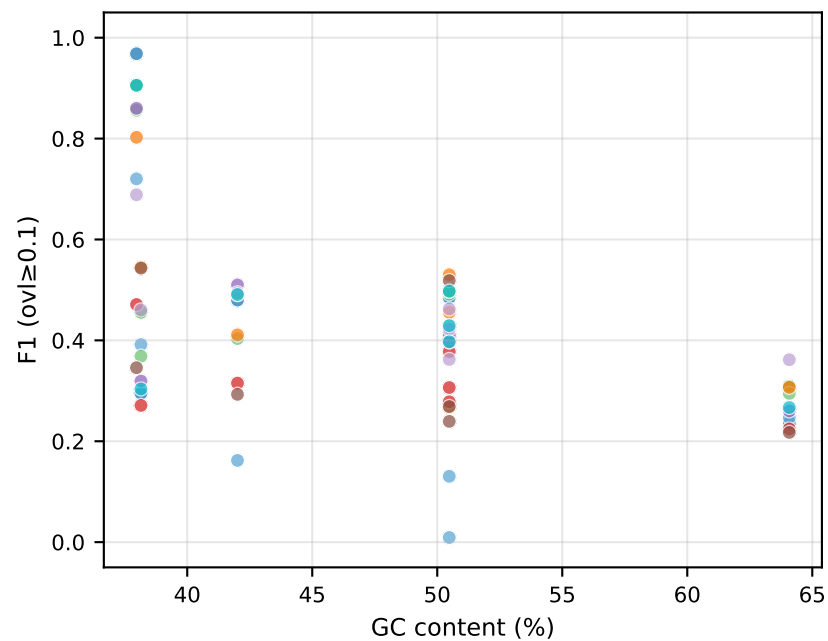
F1 delta vs default



Pareto: quality vs runtime (averaged across D1,D2,D3,D4,D6,RB_dmel,RB_ecoli)



F1 vs reference attributes — per segmenter (Setup A, $ovl \geq 0.1$)

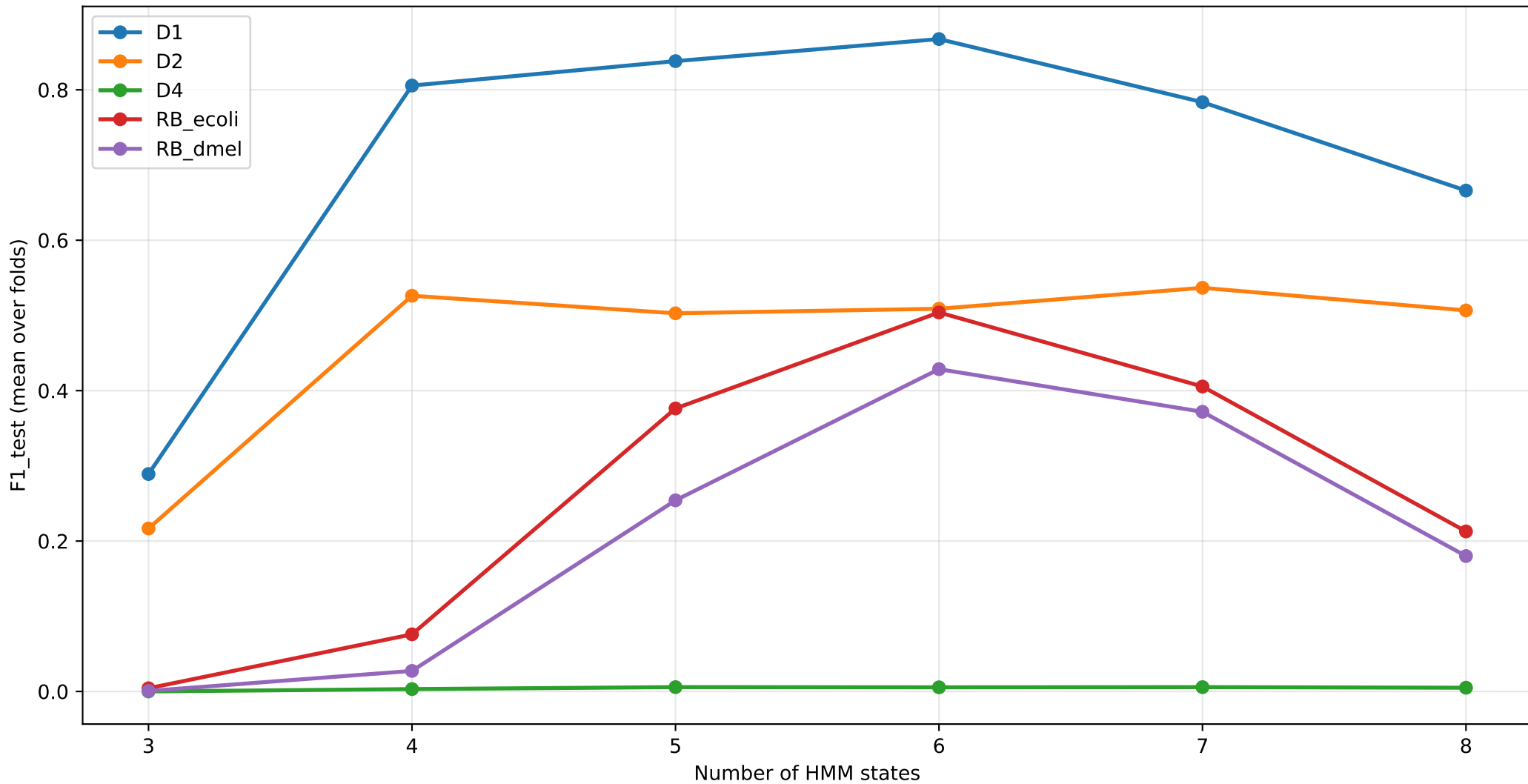


- default
- scrappie
- pelt
- binseg
- hmm
- bocd
- window
- mad
- gradient
- pelt_cuda

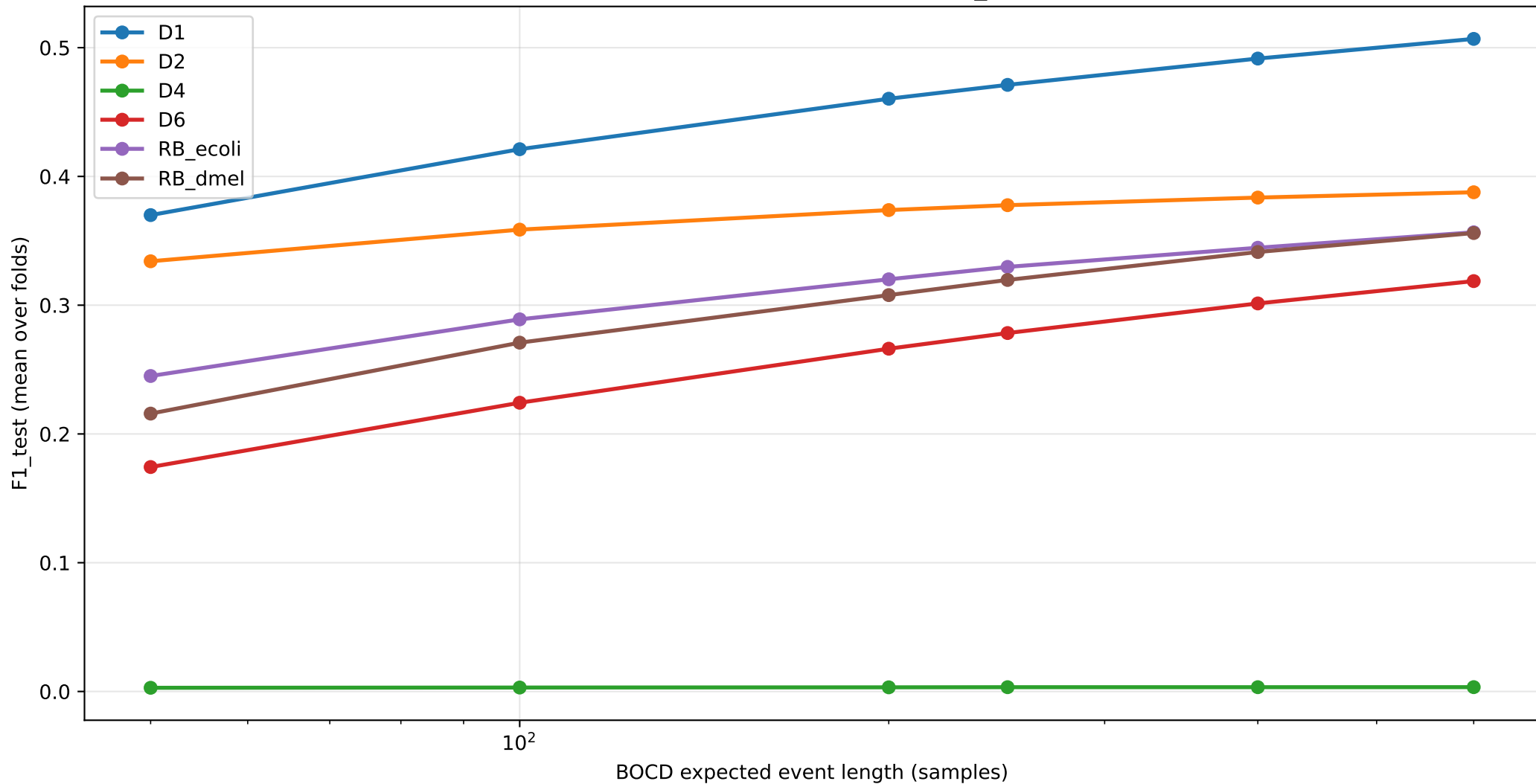
Pearson correlation: F1 (ovl $\geq$ 0.1) vs reference attribute



HMM CV sweep: F1 vs STATES (STAY=0.96)



BOCD CV sweep: F1 vs EXP_LEN



Per-dataset segmenter recommendation (BEST F1 vs FAST-near-default)

Dataset	default (F1 wall)	BEST F1 segmenter	FAST + $\geq 95\%$ F1 segmenter
D1	default (0.968 1359s)	default (0.968 1359s) [1.00×	default (0.968 1359s) [1.00×
D2	default (0.410 549s)	hmm (0.530 439s) [1.25×	hmm (0.530 439s) [1.25×
D3	default (0.295 6s)	hmm (0.545 4s) [1.52×	hmm (0.545 4s) [1.52×
D4	default (0.237 13s)	mad (0.362 83s) [0.16×	mad (0.362 83s) [0.16×
D6	default (0.484 1332s)	pelt (0.498 1337s) [1.00×	default (0.484 1332s) [1.00×
RB_dmel	default (0.480 366s)	window (0.510 6401s) [0.06×	pelt (0.491 350s) [1.05×
RB_ecoli	default (0.419 55s)	hmm (0.456 76s) [0.71×	hmm (0.456 76s) [0.71×

default vs tuned-pelt — wall-time comparison

Dataset	default wall	pelt wall	speedup	F1 default	F1 pelt
D1	1359.1s	1350.9s	1.01×	0.9682	0.9056
D2	549.1s	333.6s	1.65×	0.4097	0.4923
D3	5.8s	2.5s	2.38×	0.2951	0.4558
D4	13.0s	11.5s	1.13×	0.2366	0.3091
D6	1331.8s	1336.5s	1.00×	0.4845	0.4976
RB_dmel	366.1s	350.0s	1.05×	0.4799	0.4913
RB_ecoli	54.6s	37.3s	1.46×	0.4188	0.4292